

Trends in Technology

Why do we need to keep track of technology-performance?

↳ We need knowledge of trends so that we can design systems that can survive rapid changes in implementation technology.

↳ To be able to plan for evolution

↳ These four key implementation technologies that change at dramatic pace:

1. Micro-processors
2. Memory
3. Storage
4. Network

Micro-processors

Transistors: A switch that controls the flow of electricity.

↳ A micro-processor chip contains billions of transistors inter-connected by fine copper wires.

Circuits: A network of transistors interconnected by wires in a specific configuration to perform a specific function.

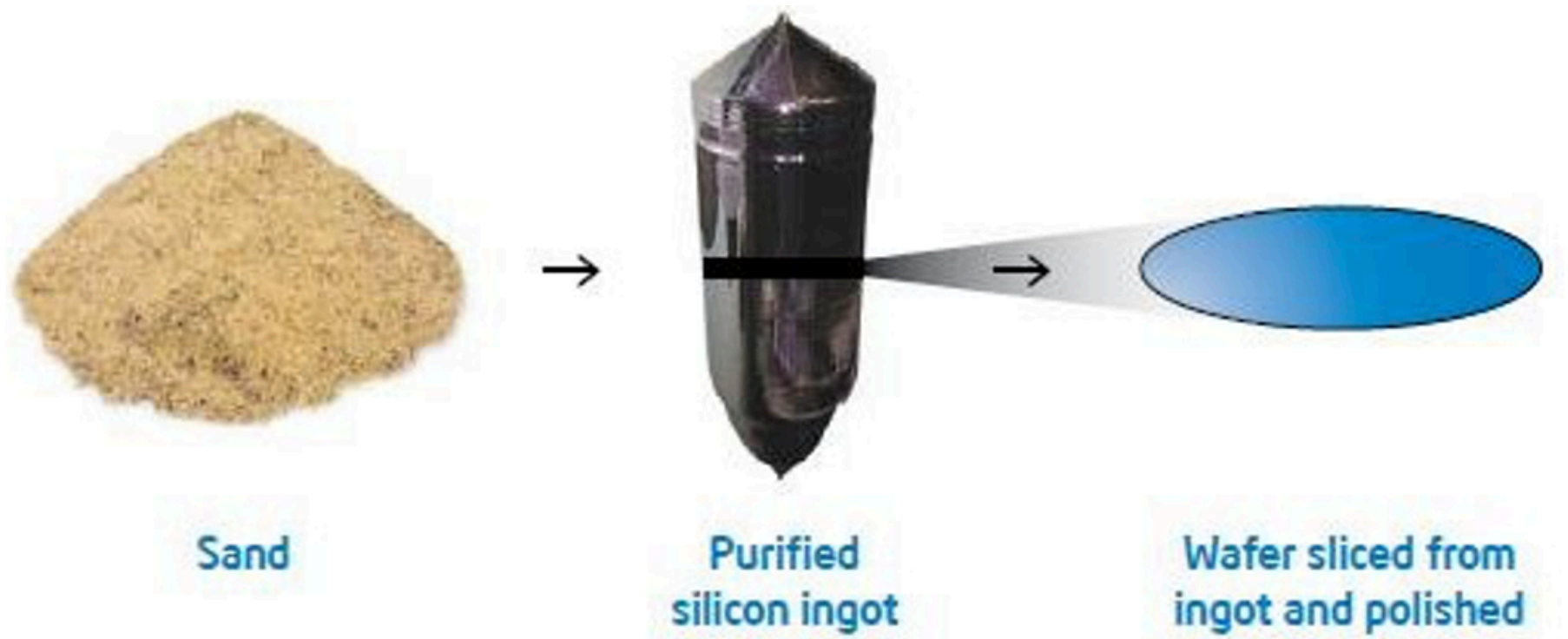
↳ Integrated circuits are characterized by feature size.

↳ feature size is the minimum size of the transistor, which decreased from 10,000nm in 1971 to 5nm in 2022.

Silicon: The principle ingredient found in common sand. It is the element used to make the wafers upon which chips are fabricated. It is the most common element on earth after oxygen.

Semi-conductor: A material that can be altered to conduct electrical current or block its passage. Silicon is a natural semi-conductor.

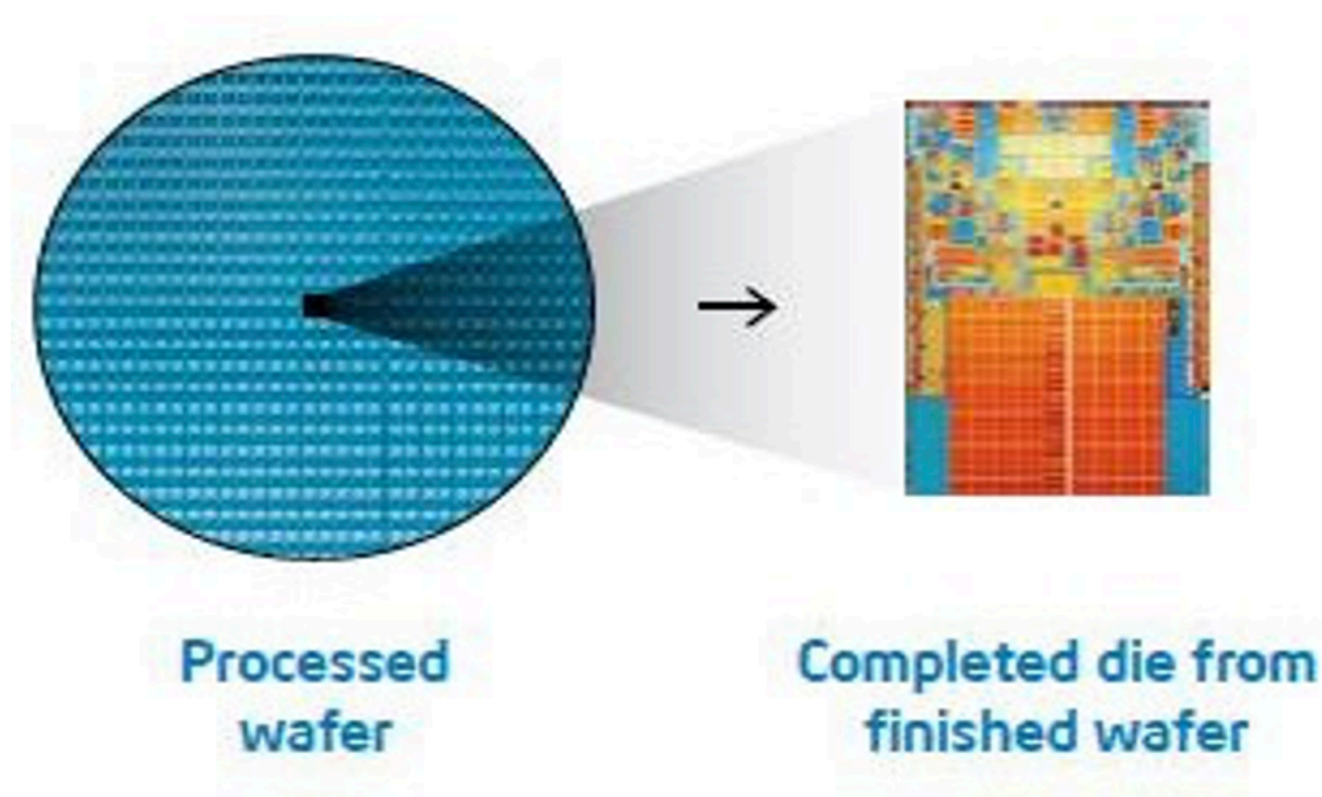
Wafer: A thin silicon disc used as the base material for building integrated circuits.



↳ Material deposited in shape of a block for convenient handling.

Die: Alternate name for chips. Dies are built in batches on wafers of silicon.

↳ A wafer is divided into batches. Each batch is called a die.



Chip: A chip is a packaged die. A thin, tiny square or rectangle that contains integrated circuitry.

Performance Indicators

Bandwidth

↳ Also called throughput

↳ The total amount of work done in a given time

OR

The rate at which data can be read from or written to

memory.

Latency:

↳ Also called response time

↳ The time between the start & the completion of an event.

Trends in Micro-processor Technology:

↳ Transistor performance increases linearly with decreasing feature size.

↳ Follows moore's law.

Microprocessor	16-Bit address/ bus, microcoded	32-Bit address/ bus, microcoded	5-Stage pipeline, on-chip I & D caches, FPU	2-Way superscalar, 64-bit bus	Out-of-order 3-way superscalar	Out-of-order superpipelined, on-chip L2 cache	Multicore OOO 4-way on chip L3 cache, Turbo
Product	Intel 80286	Intel 80386	Intel 80486	Intel Pentium	Intel Pentium Pro	Intel Pentium 4	Intel Core i7
Year	1982	1985	1989	1993	1997	2001	2015
Die size (mm ²)	47	43	81	90	308	217	122
Transistors	134,000	275,000	1,200,000	3,100,000	5,500,000	42,000,000	1,750,000,000
Processors/chip	1	1	1	1	1	1	4
Pins	68	132	168	273	387	423	1400
Latency (clocks)	6	5	5	5	10	22	14
Bus width (bits)	16	32	32	64	64	64	196
Clock rate (MHz)	12.5	16	25	66	200	1500	4000
Bandwidth (MIPS)	2	6	25	132	600	4500	64,000
Latency (ns)	320	313	200	76	50	15	4

Trends in Memory technology:

↳ Semi-conductor based DRAM (Dynamic random access memory) & SDRAM (synchronous DRAM) technology is the foundation of main memory.

- **DRAM (Dynamic RAM):** A type of memory that needs to be refreshed regularly to keep data.
- **SDRAM (Synchronous DRAM):** A type of DRAM that works in sync with the computer's clock speed, making it faster

Memory module	DRAM	Page mode DRAM	Fast page mode DRAM	Fast page mode DRAM	Synchronous DRAM	Double data rate SDRAM	DDR4 SDRAM
Module width (bits)	16	16	32	64	64	64	64
Year	1980	1983	1986	1993	1997	2000	2016
Mbits/DRAM chip	0.06	0.25	1	16	64	256	4096
Die size (mm ²)	35	45	70	130	170	204	50
Pins/DRAM chip	16	16	18	20	54	66	134
Bandwidth (MBytes/s)	13	40	160	267	640	1600	27,000
Latency (ns)	225	170	125	75	62	52	30

More efficient. Storage Technology:

Flash:

- ↳ Semi-conductor based non-volatile memory
- ↳ Electrically erased & reprogrammable
- ↳ Used in SD memory cards, USB flash drives, Solid State Drives, for standard storage in portable mobile devices
- ↳ Capacity has been doubling every 2 years
- ↳ 15 to 20 times cheaper than DRAM.

Magnetic disk

→ Amount of storage in that can be stored in a given area

- ↳ 1980s: Density doubled every 3 years
- ↳ Mid 1990s: Density doubled every 2 years
- ↳ Currently doubling in 3 years
- ↳ 15 to 20 times cheaper than flash.

Hard disk	3600 RPM	5400 RPM	7200 RPM	10,000 RPM	15,000 RPM	15,000 RPM
Product	CDC WrenI 94145-36	Seagate ST41600	Seagate ST15150	Seagate ST39102	Seagate ST373453	Seagate ST600MX0062
Year	1983	1990	1994	1998	2003	2016
Capacity (GB)	0.03	1.4	4.3	9.1	73.4	600
Disk form factor	5.25 in.	5.25 in.	3.5 in.	3.5 in.	3.5 in.	3.5 in.
Media diameter	5.25 in.	5.25 in.	3.5 in.	3.0 in.	2.5 in.	2.5 in.
Interface	ST-412	SCSI	SCSI	SCSI	SCSI	SAS
Bandwidth (MBytes/s)	0.6	4	9	24	86	250
Latency (ms)	48.3	17.1	12.7	8.8	5.7	3.6

Trends in Network Technology:

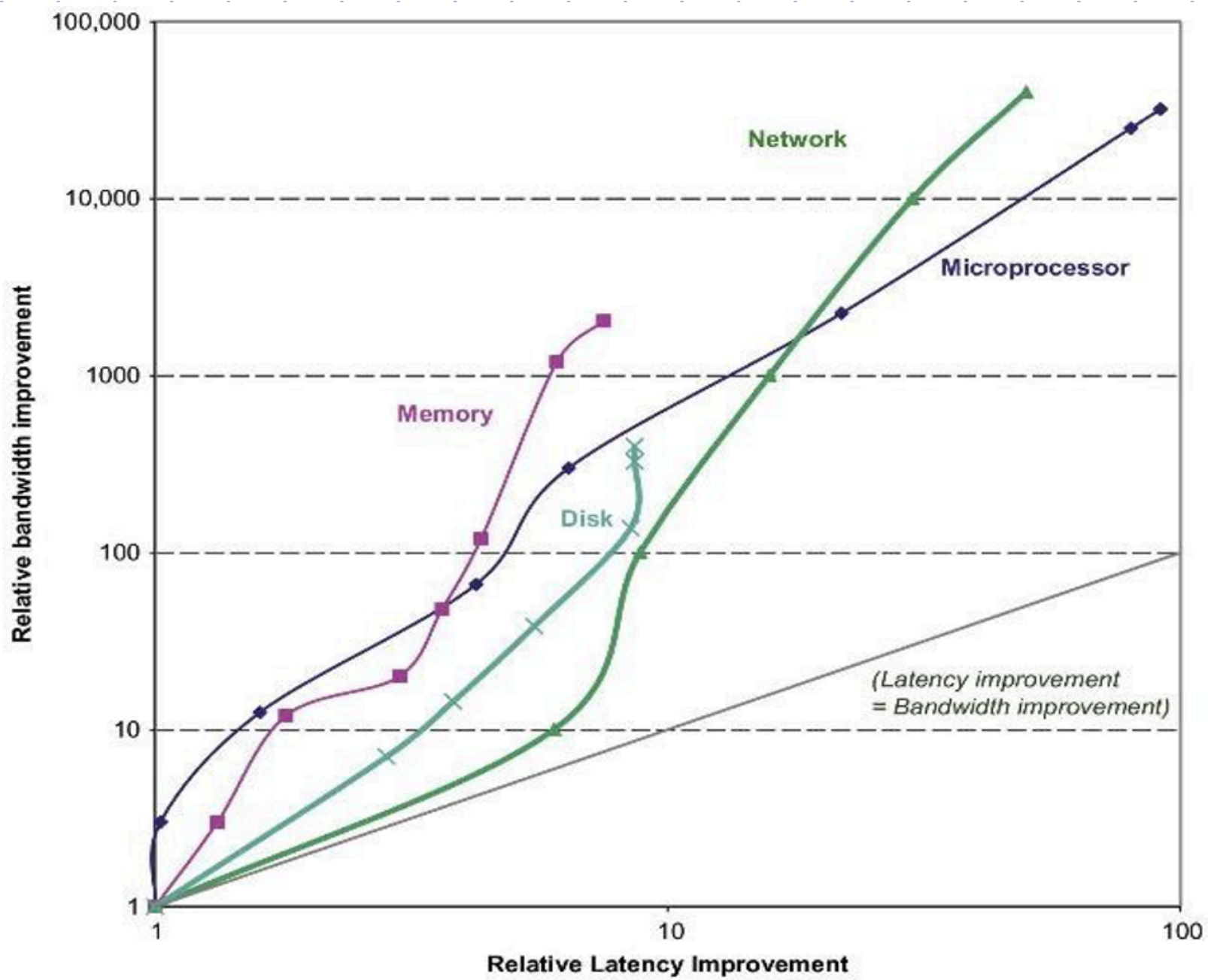
- ↳ Transmission system
- ↳ Cluster Interconnection Networks

Local area network	Ethernet	Fast Ethernet	Gigabit Ethernet	10 Gigabit Ethernet	100 Gigabit Ethernet	400 Gigabit Ethernet
IEEE standard	802.3	803.3u	802.3ab	802.3ac	802.3ba	802.3bs
Year	1978	1995	1999	2003	2010	2017
Bandwidth (Mbits/seconds)	10	100	1000	10,000	100,000	400,000
Latency (µs)	3000	500	340	190	100	60

↳ Traditionally, cost decreased at about the same rate at which density/performance increases.

↳ Designers must plan for rapid changes in technology

Bandwidth vs Latency



↳ This is for the last ~20 years

↳ A simple rule of thumb has been that bandwidth grows at least the square of the improvement in latency